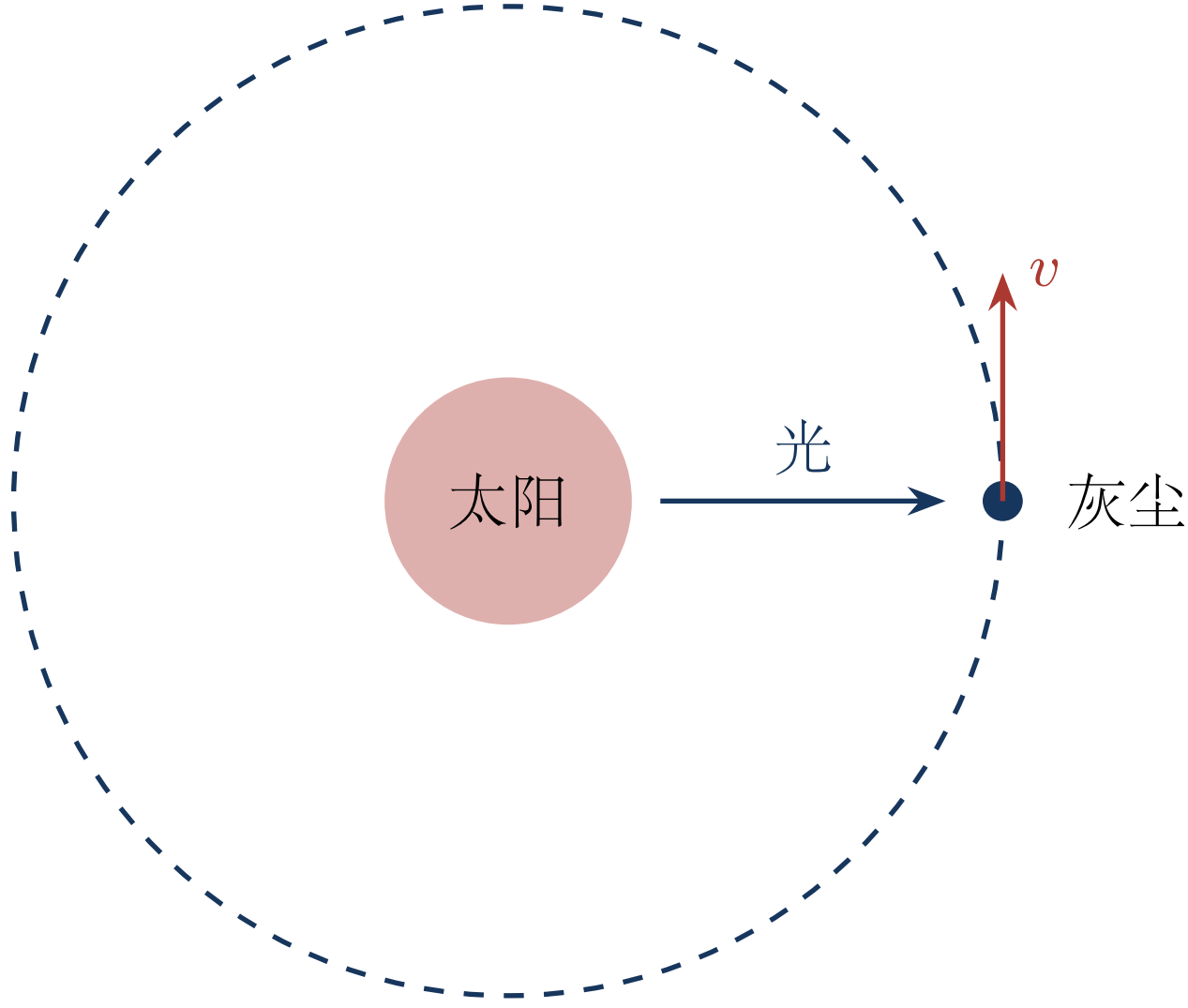
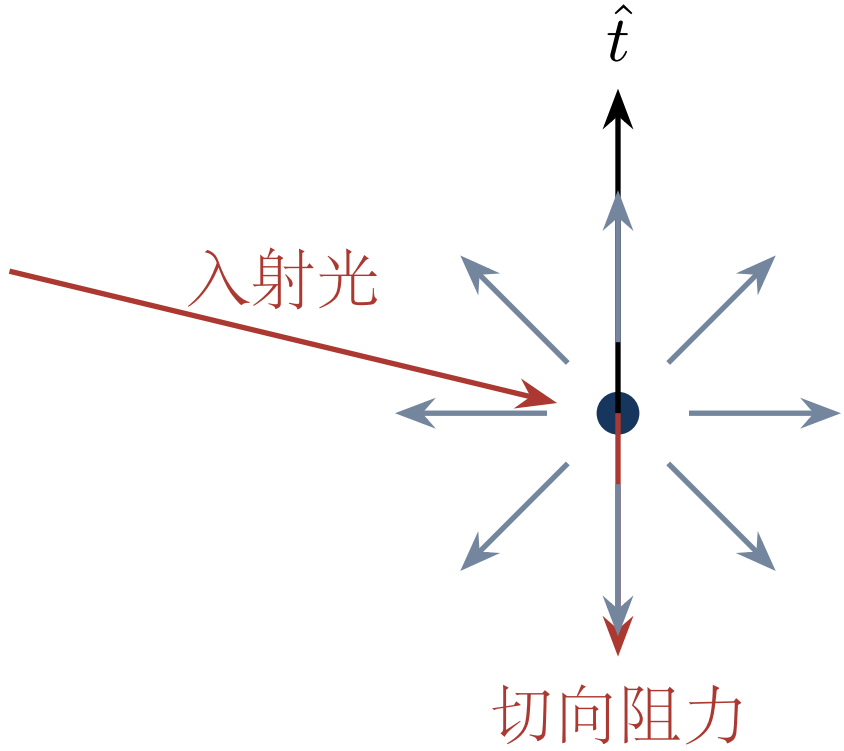
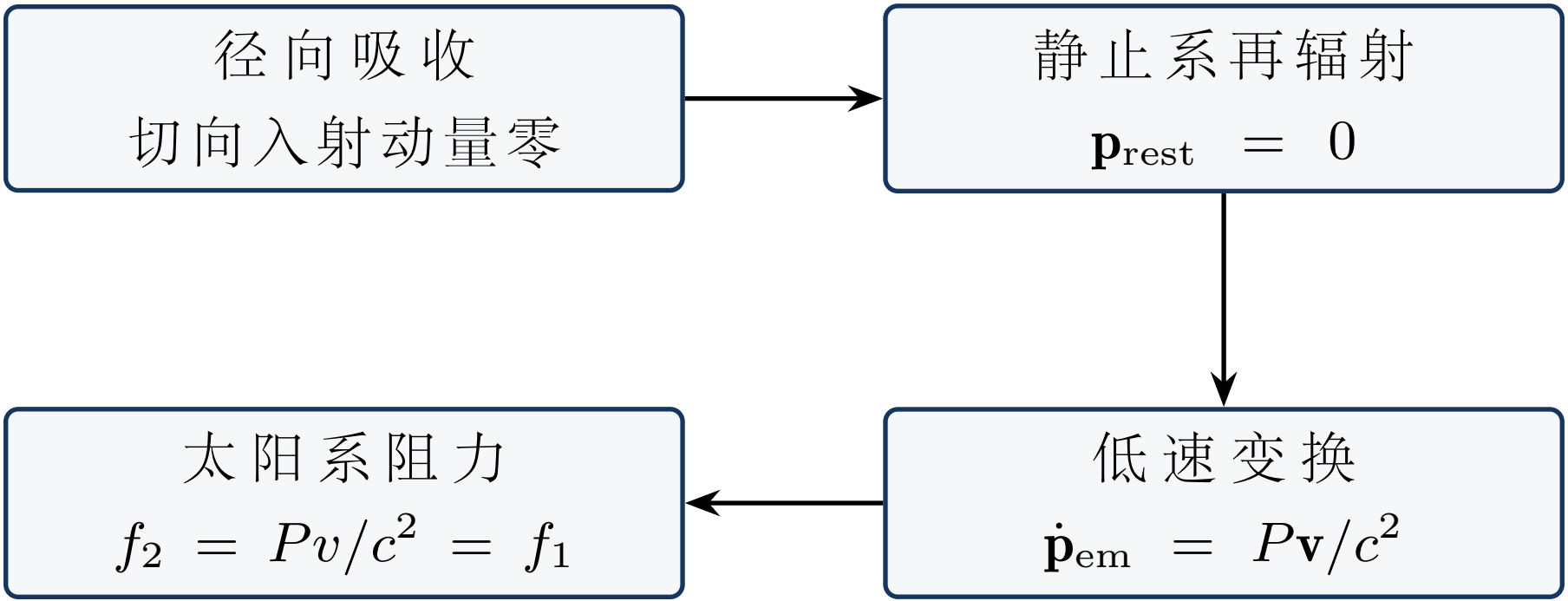


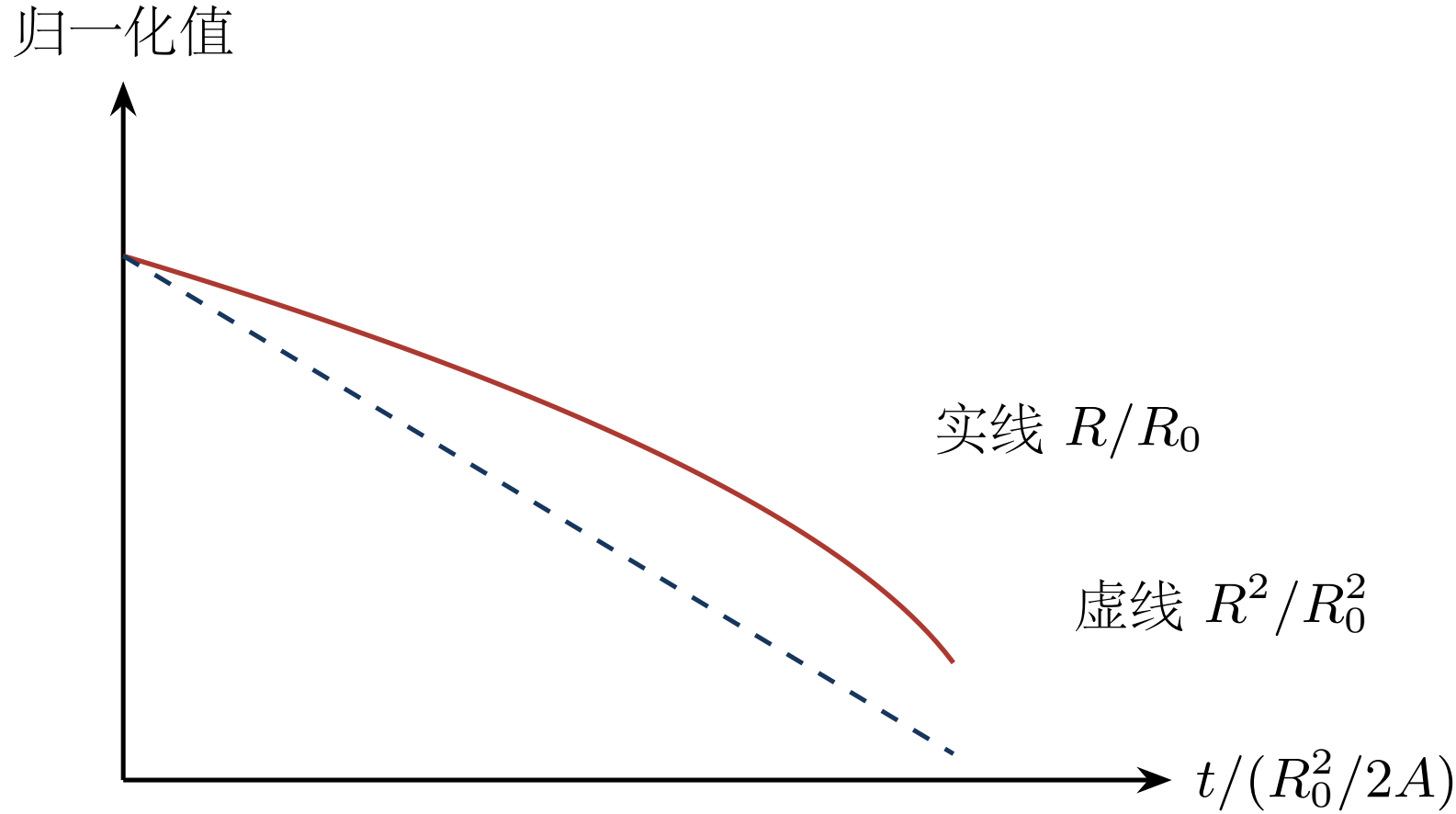
吸收截面 πr_d^2；热平衡再辐射保持近似恒定质量



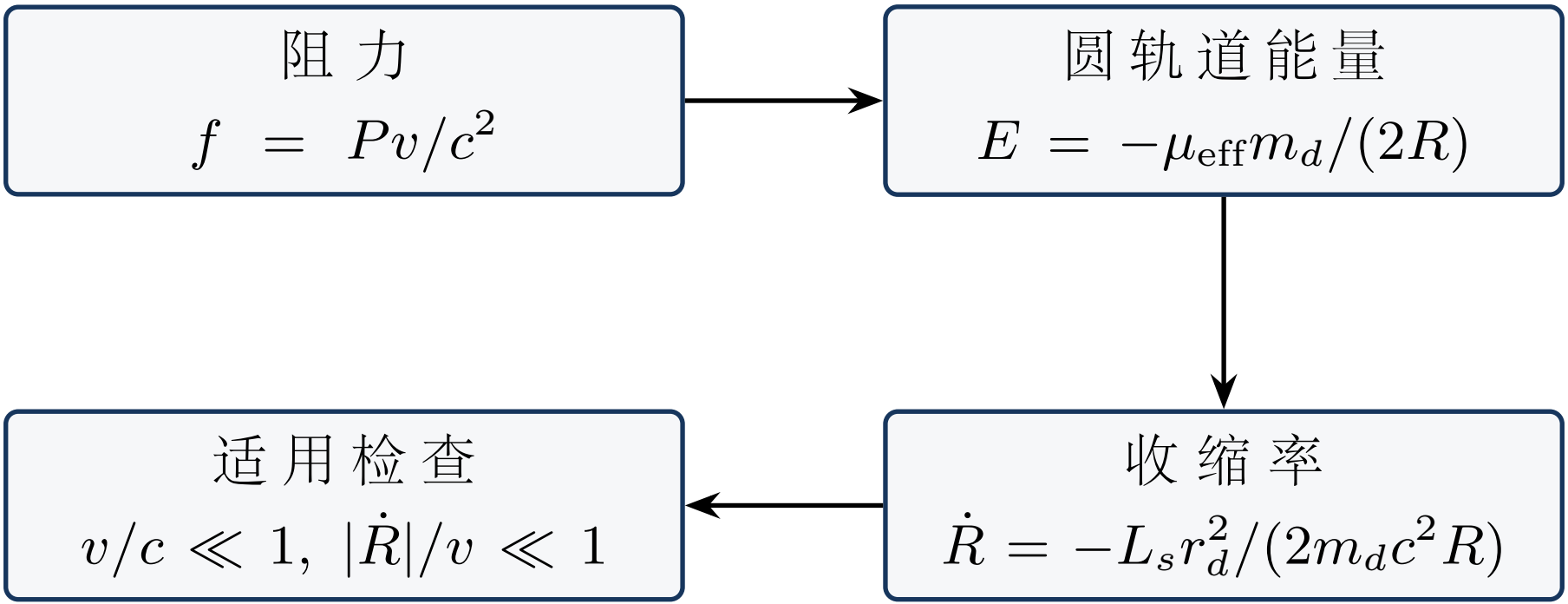
灰尘静止系：光行差角 v/c，出射总动量零



两个参考系在描述同一个净力，不叠加为双倍



$R^2 = R_0^2 - 2A_{\mathrm{PR}}t$；半径本身非线性下降



若 $\beta_{\rm rad} \geq 1$，束缚近圆轨道假设不成立